

MATH Unit 1 Question Bank — Matrices & Linear Algebra

Mathematics (MATH) · Unit 1 · Matrices & Linear Algebra · 2025

Important and frequently asked questions, grouped by marks.

Part A — Short answer (2 marks)

1. Explain Rank of a Matrix with a neat diagram and a suitable example.
2. Explain Systems of Linear Equations with a neat diagram and a suitable example.
3. Define Eigenvalues and Eigenvectors. Derive or justify the key properties involved.
4. Compare Cayley-Hamilton Theorem with Systems of Linear Equations. When would you prefer one over the other?
5. Explain Diagonalization with a neat diagram and a suitable example.
6. What is meant by Quadratic Forms? Discuss its advantages and limitations.

Part B — Long answer (10 marks)

1. Compare Rank of a Matrix with Systems of Linear Equations. When would you prefer one over the other?
2. Discuss the role of Systems of Linear Equations in Matrices & Linear Algebra.
3. Compare Eigenvalues and Eigenvectors with Diagonalization. When would you prefer one over the other?
4. What is meant by Cayley-Hamilton Theorem? Discuss its advantages and limitations.
5. Illustrate Diagonalization with a worked example.
6. What is meant by Quadratic Forms? Discuss its advantages and limitations.
7. What is meant by Rank of a Matrix? Discuss its advantages and limitations.
8. What is meant by Systems of Linear Equations? Discuss its advantages and limitations.
9. What is meant by Eigenvalues and Eigenvectors? Discuss its advantages and limitations.
10. Compare Cayley-Hamilton Theorem with Eigenvalues and Eigenvectors. When would you prefer one over the other?
11. Illustrate Diagonalization with a worked example.
12. Write short notes on Quadratic Forms.